A Rule-Based Detector for Swedish Translated from English

Technical report: design, calibration and evaluation on Europarl

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Abstract

We describe a rule-based expert system that classifies Swedish text as written in Swedish or translated from English. It consists of 49 hand-written rules over regular expressions, word lists and counts, grouped into seven families. Each rule contributes a log-likelihood ratio whose parameters were typed by hand from training-split statistics, and every verdict comes with span-level evidence. On Europarl v7 with a date-based split and topic words masked, the engine reaches accuracy / ROC-AUC of .613 / .666 on single sentences, .747 / .833 on 100-token chunks and .920 / .977 on 500-token chunks of the held-out validation period (2004–2006), against an AUC of .583, .671 and .792 for a sentence-length baseline. Logistic regression and gradient-boosted trees over the same expert features add between .006 and .036 AUC. Several literature claims about translationese, among them explicitation through connectives, do not hold in this corpus.

1 Introduction

Translationese, the set of statistical properties that distinguish translated text from text originally written in the same language, was named in a study of exactly this language pair: Swedish novels translated from English (Gellerstam 1986). Computational work has since shown that translated text can be separated from original text with high accuracy on long chunks (Baroni and Bernardini 2006; Koppel and Ordan 2011; Volansky et al. 2015), mostly with function-word, character and part-of-speech n-gram features in supervised classifiers. For Swedish Europarl, the only classifier we are aware of is a three-class SVM (original Swedish, from English, from other languages) over POS and character n-grams on about 2,000-token chunks, which reached 94.1% in cross-validation and 80% on a held-out set (Joelsson 2016).

This report describes a system built under a different constraint: it is fully rule-based. It uses tokenisation, regular expressions, word lists, counts and hand-written rules with hand-set parameters, and no trained model, tagger or parser. The rules implement features from the translation-studies literature (interference, normalisation, unique items, explicitation) and from contrastive Swedish–English grammar, and each verdict is explained by the rules that fired and the words they fired on. We evaluate it on the Swedish side of Europarl with a date-based split, measure how much an ML model on top of the same features gains, and report which literature claims hold and which do not in this corpus.

The project's literature review (`docs/literature.md`) records how each cited source was checked. Some were confirmed only through citing sources, among them Gellerstam (1986), Baker (1993) and Tirkkonen-Condit (2004); claims attributed to them here follow those secondary accounts.

2 Task and data

Corpus and labels. The data is the Swedish side of Europarl v7 (Koehn 2005). Label 0 marks a speech given in Swedish, label 1 a speech given in English and translated into Swedish by the European Parliament. Europarl records a speech's source language mainly in the *other* language versions: a naive reading of the `LANGUAGE` attribute in the Swedish files finds only 154 original Swedish turns. Labels therefore combine the attribute in `txt/sv` and `txt/en` (same file, same speaker ID) with inline `(XX)` language markers. A turn is dropped when these sources disagree, when the Swedish and English speaker names disagree, or when the text does not look Swedish. Turns in other source languages and untagged turns are dropped as well. Of 196,477 speaker turns, 5,994 original and 34,149 translated turns are kept. Tag inconsistency across language files is a known source of label noise in Europarl (Ustaszewski 2019).

Table 1: Date-based splits (balanced files; counts per class, so each set has twice as many units). Sessions: first and last session date present and number of sessions. Tokens: whitespace tokens of the balanced 500-token file, both classes. Source: `data/MANIFEST.md`.

split	sessions	n	sentences	100-token	500-token	tokens (500)
trn	1997-01-14 – 2004-05-05	483	43 952	7 085	1 367	1 400 951
val	2004-07-20 – 2006-12-18	140	10 864	1 677	318	325 339
test	2007-01-15 – 2011-07-06	250	19 332	3 058	589	603 493

Units. Three granularities are built: sentences of at least three tokens; and chunks of consecutive sentences of one class within one session date, emitted as soon as they reach 100 or 500 whitespace tokens (leftovers are dropped). The translated class is several times larger than the original class, so balanced files downsample it with a fixed seed. All figures in this report are on balanced files.

Split and protocol. The split is by session date, so a session never straddles a cut (Table 1). Each split also falls into a different period: trn covers 1997 to mid-2004, val the following parliamentary term after the 2004 enlargement, test 2007–2011. Speakers overlap little: of 151 distinct original-language speakers in trn and 89 in val, 10 occur in both. The protocol is enforced by the tools:

- **trn** is the only split whose texts are read. Rule values are read off trn statistics, and examples, keyword-in-context lines and errors are inspected only on trn.
- **val** is used for aggregate metrics only, a few runs per change. The calibration tools refuse it, and error listing is disabled.
- **test** is locked for one final run and has **not been run** for this report.

Topic masking. Country, nationality, city, party and politician tokens can be replaced by a placeholder before scoring. Every result is reported masked and unmasked. A deliberately topic-keyed baseline must fall to chance under masking, which checks that the mask removes the topical signal. Masked figures are reported throughout; for the engine they differ from the unmasked ones by at most .001.

3 Method

3.1 Engine

Text processing. Input is normalised (quotation marks, dashes, ellipses, whitespace, soft hyphens, Unicode NFC) with a character offset map back to the original input. A rule-based sentence splitter ends sentences at `.`, `!`, `?` and `...` before an upper-case letter or digit, except after known abbreviations and initials; *Herr talman!* 'Mr President!' is its own sentence, and vocative-only sentences are flagged so that sentence-position and sentence-length rules can skip them. The tokeniser keeps abbreviations (*t.ex.*, *bl.a.*), numbers with decimal comma or space separators and hyphenated compounds (*EU-länderna*) as single tokens. Rules read only from a lazily cached document object (sentences, tokens, lower-cased words, sentence-initial words, punctuation counts); regular expressions run on the lower-cased normalised text.

Rule types. Every rule returns a log-likelihood ratio in nats, $\mathrm{LLR} = \ln P(\text{text} \mid T)/P(\text{text} \mid O)$, so that positive values point to *translated*. Five types cover all rules:

- **HitRule**, for rare, high-precision items such as calques: a fixed LLR per match (or one per lexicon entry), $\mathrm{LLR} = \sum_{\text{hits}} \ell_h$. Absence is no evidence.
- **RateRule**, the frequency of an item relative to length under a Poisson model, with hand-set rates r_O, r_T per 1000 words. For k hits in n words,

$$\mathrm{LLR} = k \ln \frac{r_T}{r_O} - n \frac{r_T - r_O}{1000}.$$

  The second term is evidence from absence and grows with length.
- **ChoiceRule**, which of two competing variants was chosen (binomial), with p_O, p_T the share of variant A among A and B in each class:

$$\mathrm{LLR} = k_A \ln \frac{p_T}{p_O} + k_B \ln \frac{1 - p_T}{1 - p_O}.$$

  Conditioning on a choice having been made makes it the most robust type to length and topic, but it discards information when the rate of variant B also differs between the classes.

- **StatRule**, a per-unit statistic (e.g. words per sentence) modelled as Gaussian per class; with a pooled SD the LLR is linear in the mean and decomposes into per-sentence shares. The statistics family uses a variant whose SD of the text mean is $\sqrt{\sigma^2/n + \tau^2}$, where τ is between-text variation that does not shrink with length, so the evidence saturates on long texts.
- **PredicateRule**, arbitrary code returning an LLR, used for the word-order rules (clause opening classification over closed word lists for finite verbs, subject pronouns and fronted adverbials).

Combination. Each rule's LLR is multiplied by a weight w_r (default 1) and clipped to $\pm c_r$. Rule contributions are summed per family, the family sum is clipped to $\pm C_f$, and the score is the sum over families:

$$s = \sum_f \mathrm{clip}\Big(\sum_{r \in f} \mathrm{clip}(w_r \,\mathrm{LLR}_r,\ \pm c_r),\ \pm C_f\Big), \qquad P(\text{translated}) = \frac{1}{1 + e^{-s}}.$$

The prior is 0 (balanced classes) and the decision threshold is $s > 0$. Family caps (2 to 6) bound the damage of correlated rules that read the same tokens.

Explanations. Besides the score, the engine reports each rule's contribution (with a pseudo-entry for the amount a family cap removed, so the contributions sum to the score), a human-readable evidence sentence per rule, and character spans for every piece of evidence, mapped back to the original input. Each span carries its signed share of the rule's contribution; what is not tied to a span (the absence term of rate rules, residuals of statistics) is reported at document level. Section 5 shows this output.

3.2 Calibration protocol

Rule values are set by hand, as a knowledge engineer would. A rule's pattern and its rationale are written before any numbers are looked at. A helper tool then prints per-class statistics on trn (by default the balanced 500-token file, about 0.7M words per class): hits, rate per 1000 words, share of texts with a hit, variant shares for choices, random keyword-in-context lines, the share of hits from the five most frequent speakers, and per-year rates. The KWIC lines are read to check that the pattern matches what was meant, and the rounded values (at most two significant digits) are typed into the rule with a comment recording what was seen. There is no automated fitting and no search over weights, caps or thresholds. Further constraints: every rule cites a literature entry or is marked as expert judgement; names of countries, parties, persons, cities and institutions are banned from patterns; every rule is checked for speaker concentration (the share of hits from the five most frequent speakers); and drifting items get small caps. Values were set on trn, and val was scored a handful of times to compare rule sets.

3.3 Integration of the families

The seven families were calibrated separately and then combined, so evidence that two families read off the same tokens was initially counted twice. To find such overlaps, every trn text was scored with all rules and two diagnostics were computed: the within-class Pearson correlation of every pair of rule contributions (separately in O and T, on 14,170 100-token and 2,734 500-token chunks), and the character overlap of their evidence spans. Pairs with same-direction $|r| > .3$ or shared spans were resolved by hand, by removing the overlapping pattern from one rule, disabling one rule, or halving one weight. For example, a register-family rate of *av* correlated at $r = .96$ (O) and .94 (T) with the statistics family's *av* rule and was disabled, and sentence-initial *Därför*, *Dessutom* were removed from the connective rule because the word-order family scores the same tokens as a topicalised opening. Of 60 defined rules, 49 are enabled. On trn (masked), resolution moved 100-token AUC from .902 to .907. Family caps bind almost only on 500-token chunks (particles on 20.0% of original chunks, calques on 11.7% of translated ones), the clipped amount points to the true class in 90–100% of these cases, and lifting any one cap changes trn accuracy by 0 and AUC by at most .0002; no cap was changed.

4 Features

Table 2 lists the main rules with their hand-set class parameters; all 49 rules are in Appendix A. The families are:

- **starter**: modal particles (*ju, nog, väl, visst*) and words per sentence.
- **connectives and punctuation**: native connective adverbs and additive *också* (O), Swedish stock equivalents of English discourse markers (*i synnerhet* 'in particular', *i själva verket* 'in fact'), sentence-initial *Och/Men*, a comma after a fronted adverbial (*Dessutom, vi måste* 'Moreover, we must'), comma habits

around clausal conjunctions, dropped complementiser *att* and relative *som*, written-out abbreviations, dash-delimited asides, parentheses and exclamation marks.

- **register**: lexical choices between native variants, e.g. *kommer att* vs *ska/skall* for the future, *skulle vilja* vs *vill*, *tycka* vs *anse*, *väldigt* vs *mycket*, and rates of *bland annat*, *samt*, evaluative *bra*, *självklart* and *den/det/de här*.
- **particles and pronouns**: clause-internal particles (*nämligen, ändå, alltså*, anaphoric *då*), stance adverbs, Swedish particle verbs (*slå fast*, *lyfta fram*), generic *man*, copular *det är*, possessive and personal pronouns, *ni* vs *man*.
- **calques**: a lexicon of 43 English calques with per-entry LLRs (*det faktum att* 'the fact that', *när det rör* 'as regards', *de av oss som* 'those of us who', *i det förflutna* 'in the past'), two native counterparts, *försöka att* + infinitive, and pied-piped relatives (*i vilket* 'in which').
- **word order**: a heuristic classifier of main-clause openings (topicalised adverb, object or clause with V2 inversion vs subject-first), and English order leaking into Swedish (no inversion after a fronted adverbial, V3, main-clause negation order in subordinate clauses).
- **statistics**: the share of long words (the LIX criterion) and the share of *av* 'of, by' after nominalisations and elsewhere; these abstain below 50 words.

Table 2: Main rules by family. Dir: the direction a hit points to (O original, T translated). Hand-set values are typed from trn statistics (O / T): rate rules in hits per 1000 words, choice rules as the share of variant A (the first-named variant), stat rules as class means (words per sentence; *av* per 1000 words); for the hit rules, log-likelihood ratio per hit. AUC: the rule's contribution alone as a score on trn 100-token units; ×LR: multiplier on the hand-set contribution learned by a logistic regression over all 49 rule contributions on trn (Section 6.3). Full list in Appendix A.

rule	dir	feature	hand-set O / T	AUC	×LR
starter (family cap 6)					
`modal_particles`	O	modal particles *ju, nog, väl, visst*	.92 / .092	.591	0.802
`sent_len`	T	words per sentence	18 / 21	.675	1.379
connectives & punctuation (family cap 4)					
`native_connectives`	O	middle-field *dessutom, därför, dock, därmed, inte minst*	4.7 / 1.9	.640	1.904
`additive_ocksa`	O	additive *också* 'also'	5.5 / 2.2	.641	2.067
`calque_connectives`	T	*i synnerhet, i själva verket, till sist* 'in particular, in fact, finally'	.34 / 1.5	.520	0.866
`initial_och`	T	sentence-initial *Och* 'And'	.016 / .17	.465	1.419
`initial_adverbial_comma`	T	comma after a fronted adverbial: *Dessutom, vi …*	P(A) .01 / .091	.569	0.678
`clause_comma_omission`	T	no comma before clausal *men, eftersom, medan, vilket*	P(A) .043 / .17	.537	0.812
`written_out_abbrev`	T	*till exempel* vs *t.ex.*, *så kallad* vs *s.k.*	P(A) .28 / .71	.547	1.529
register (family cap 6)					
`kommer_att_vs_ska`	T	future *kommer att* vs *ska/skall*	P(A) .24 / .6	.667	0.713
`skulle_vilja_vs_vill`	T	*jag/vi skulle vilja* vs *jag/vi vill* 'would like to' vs 'want to'	P(A) .097 / .32	.559	0.838
`tycker_vs_anser`	O	opinion verb *tycka* vs *anse* 'think, consider'	P(A) .58 / .2	.550	0.395
`valdigt_vs_mycket`	O	*väldigt* vs *mycket* + adjective 'very'	P(A) .43 / .1	.539	0.603
`bland_annat`	O	*bland annat, bl.a.* 'among other things'	.64 / .021	.568	1.408
`samt`	O	coordinator *samt* 'and, as well as'	.85 / .25	.567	1.244
`dem_har_rate`	T	demonstrative *den/det/de här* 'this, these'	.17 / .81	.492	2.017
particles & pronouns (family cap 4)					
`namligen`	O	clause-internal *nämligen* 'you see, namely'	.42 / .02	.563	1.287
`stance`	O	*egentligen, visserligen, i och för sig, i alla fall*	.51 / .12	.560	0.854
`generic_man`	O	generic *man* 'one'	3.5 / 2.4	.592	1.293
`ni_vs_man`	T	*ni* 'you (pl.)' vs generic *man*	P(A) .17 / .35	.574	0.455
`poss_3rd`	T	*hans, hennes, deras, dess* 'his, her, their, its'	.98 / 2.8	.538	0.736
calques (family cap 3)					
`lexicon`	T	43 calques, e.g. *det faktum att* 'the fact that', *när det rör* 'as regards'	+0.5 to +1.5 per hit	.603	1.460
`pied_piped_relative`	T	*i vilket*, *för vilka* 'in which, for which'	+0.8 per hit	.521	0.724
word order (family cap 4)					
`fronted_theme`	O	topicalised non-subject + V2: *Därför anser jag*, *Det vet vi*	per-sentence LLRs	.637	0.954
`subject_theme`	T	subject in the prefield: pronoun, NP, *det*, cleft	per-sentence LLRs	.532	0.748
`v2_violation`	T	no inversion after a fronted adverbial, V3, embedded negation	per-sentence LLRs	.502	0.922
statistics (family cap 2)					
`av_other`	T	share of *av* 'of, by' outside nominalisations	13 / 17	.594	0.707

5 Annotated examples

The boxes below show the engine's output on six units from **trn**; no val or test text is shown anywhere in this report. Outcomes use translated as the positive class (TP, TN, FP, FN). Spans are coloured by the net direction of the evidence covering each character (toward translated, toward original), with intensity growing with the summed contribution share; overlapping spans are merged per character. A superscript after a span gives the number of its rule in the list below the text. Sentence-level statistics (words per sentence, word length, *av* shares) have no token spans and appear only in the list. The lists and highlights are generated at build time from the current engine; the notes are the analyst's reading and are checked against the engine's contributions in the build. Sentence examples carry an English gloss.

Example 1 sentence **TP** true: translated P(translated) = 0.992 score +4.81 24 words

Till sist[3],[1] jag var[2] både glad och smickrad när president Chirac hänvisade till detta betänkande i sitt tal till parlamentet för några veckor sedan.

'Finally, I was both pleased and flattered when President Chirac referred to this report in his speech to Parliament a few weeks ago.'

1	`connpunct.initial_adverbial_comma`	+2.00	comma after a sentence-initial adverbial ('Dessutom, …')
2	`wordorder.v2_violation`	+1.50	English word order leaking into Swedish (V3 / no inversion / embedded negation)
3	`connpunct.calque_connectives`	+1.17	Swedish equivalents of English discourse markers (i synnerhet, i själva verket …)

28 other rules: +0.15 in total (mostly absence and length terms).

Note. *Till sist, jag var* follows English *Finally, I was.* Three rules read three properties of the same construction: the comma after the fronted adverbial (punctuation, at its rule cap of 2), the missing V2 inversion (word order) and the structuring conjunct *till sist* 'finally' (lexis).

Example 2 sentence **TN** true: original P(translated) = 0.014 score −4.27 20 words

Personligen[4] tycker[2] jag[3] nämligen[1] att kampanjen för att förhindra våld mot kvinnor faktiskt är viktigare för framtiden än denna världsutställning.

'Personally, you see, I think that the campaign to prevent violence against women is actually more important for the future than this world exhibition.'

1	`particles.namligen`	−2.50	clause-internal particle 'nämligen' (explanatory 'you see')
2	`register.tycker_vs_anser`	−1.06	opinion verb 'tycka' vs 'anse'
3	`wordorder.fronted_theme`	−0.67	topicalised non-subject in the prefield (V2 inversion): which kind
4	`connpunct.initial_adverbial_comma`	−0.09	comma after a sentence-initial adverbial ('Dessutom, …')

28 other rules: +0.05 in total (mostly absence and length terms).

Note. Three native choices without an English trigger: the clause-internal particle *nämligen* 'you see' (at its rule cap of 2.5), *tycker* rather than *anser* 'think', and the fronted adverb with V2 inversion *Personligen tycker jag* 'personally think I'.

Example 3 sentence **FP** true: original P(translated) = 0.925 score +2.52 14 words

Till sist[2],[1] för att detta arbete skall[3] bli framgångsrikt får kommissionen inte lämnas ensam.

'Finally, for this work to be successful, the Commission must not be left alone.'

1	`connpunct.initial_adverbial_comma`	+2.00	comma after a sentence-initial adverbial ('Dessutom, …')
2	`connpunct.calque_connectives`	+1.17	Swedish equivalents of English discourse markers (i synnerhet, i själva verket …)
3	`register.kommer_att_vs_ska`	−0.64	future/intention auxiliary 'kommer att' vs 'ska/skall'

28 other rules: −0.02 in total (mostly absence and length terms).

Note. An original sentence that opens like Example 1, *Till sist,* with a comma. The comma rule (at its cap) and *till sist* outweigh *skall*, which counts as the native side of the *kommer att* vs *ska/skall* choice. Here the comma precedes a fronted purpose clause (*för att …*), where it is ordinary Swedish punctuation: a false positive of the comma rule.

Example 4 sentence **FN** true: translated P(translated) = 0.039 score −3.21 21 words

Då[4] tänkte vi[3] att det kanske rent ekonomiskt kunde bli bra[2] i Hongkong men att det politiskt skulle gå väldigt[1] dåligt.

'Then we thought that it might go well in Hong Kong in purely economic terms, but that politically it would go very badly.'

1 `register.valdigt_vs_mycket`	−1.46	degree adverb 'väldigt' vs 'mycket' + adjective
2 `register.bra`	−1.06	evaluative 'bra'
3 `wordorder.fronted_theme`	−0.67	topicalised non-subject in the prefield (V2 inversion): which kind
4 `connpunct.initial_adverbial_comma`	−0.09	comma after a sentence-initial adverbial ('Dessutom, …')

28 other rules: +0.06 in total (mostly absence and length terms).

Note. A translation with three native-register choices: *väldigt* rather than *mycket* before an adjective, evaluative *bra* 'good, well', and V2 inversion after the fronted adverb (*Då tänkte vi* 'then thought we'). Apart from small absence and length terms, no evidence points to translation.

Example 5 100-token chunk **TP** true: translated P(translated) = 1.000 score +9.48 117 words

En uppfördandekodex kanske inte räcker. Slutligen,[3] när det gäller turism, skulle jag vilja[4] gratulera samtliga föredragande, och i synnerhet[2] Torres Marques för hennes[5] betänkande. Turism är[8] Europas och världens största industri, trots det har vi i Europa förlorat 10 procent av vår[9] marknadsandel sedan 1960. Turism en klassisk industri på den inre marknaden. Eurons födelse kommer[8] att[1] ge denna marknad en oerhörd knuff framåt, och hela turistindustrin i allmänhet[2]. Jag tror[8] att vi kommer att[1] få tre typer av vinnare genom introducerandet av euron. Först och främst[2,3] de som arbetar inom industrin. Euron kommer[8] inte bara att skapa fler arbetstillfällen, den kommer att[1] ge större möjlighet till överblick av löner, vilka i sin tur kommer att[1] förbättras.

1 `register.kommer_att_vs_ska`	+2.50	future/intention auxiliary 'kommer att' vs 'ska/skall'
2 `connpunct.calque_connectives`	+2.00	Swedish equivalents of English discourse markers (i synnerhet, i själva verket …)
3 `connpunct.initial_adverbial_comma`	+2.00	comma after a sentence-initial adverbial ('Dessutom, …')
4 `register.skulle_vilja_vs_vill`	+1.19	'jag/vi skulle vilja' vs 'jag/vi vill'
5 `particles.poss_3rd`	+0.67	3rd-person non-reflexive possessives 'hans/hennes/deras/dess'
6 `starter.sent_len`	−0.48	words per sentence *(no span)*
7 `stats.av_nominal`	+0.43	'av' after a definite nominalisation ('genomförandet av') *(no span)*
8 `wordorder.subject_theme`	+0.22	subject in the prefield: pronoun / NP / det / demonstrative / cleft
9 `particles.poss_1st`	+0.12	1st-person possessive determiners 'min/mitt/mina/vår/vårt/våra'

27 other rules: +1.03 in total (mostly absence and length terms); family cap `connpunct`: −0.20.

Note. Evidence from several families: future *kommer att* with no *ska/skall* (the choice rule at its cap of 2.5), *Slutligen,* with a comma, the structuring conjuncts *i synnerhet* 'in particular' and *i allmänhet* 'in general', *skulle jag vilja* 'I would like to', and the possessive *hennes* 'her'. The connectives-and-punctuation family exceeds its cap of 4 and is clipped.

Example 6 100-token chunk **FP** true: original P(translated) = 0.986 score +4.24 101 words

Det gäller[9] betänkandet av Florenz som också[6] har debatterats under detta sammanträde. Låt mig understryka att kommissionen kommer att[1] säkerställa[2] att det förslag vi kommer[4] att[1] lägga fram tidigt nästa år om oktaBDE och dekaBDE kommer att[1] vara förenligt med vårt[8] förslag om begränsning av farliga ämnen i elektrisk och elektronisk utrustning. Eftersom riskbedömningen när det gäller oktaBDE och dekaBDE är långt framskriden blir det, om så skulle erfordras, möjligt för kommissionen att vidta åtgärder mycket tidigare än 2006. Då[10] riskbedömningarna slutförts och tillgången på säkra ersättningsprodukter undersökts, bör det bli möjligt att fastställa i vilka fall som användningen innebär risker.

1 `register.kommer_att_vs_ska`	+2.50	future/intention auxiliary 'kommer att' vs 'ska/skall'
2 `calques.lexicon`	+0.60	English calques and anglicisms
3 `stats.av_nominal`	+0.45	'av' after a definite nominalisation ('genomförandet av') *(no span)*
4 `connpunct.som_drop`	+0.43	object relative without 'som' ('det arbete ni har gjort')
5 `starter.sent_len`	+0.29	words per sentence *(no span)*
6 `connpunct.additive_ocksa`	−0.23	additive 'också'
7 `particles.poss_3rd`	−0.15	3rd-person non-reflexive possessives 'hans/hennes/deras/dess' *(no span)*
8 `particles.poss_1st`	+0.14	1st-person possessive determiners 'min/mitt/mina/vår/vårt/våra'
9 `wordorder.subject_theme`	−0.11	subject in the prefield: pronoun / NP / det / demonstrative / cleft
10 `connpunct.initial_adverbial_comma`	−0.09	comma after a sentence-initial adverbial ('Dessutom, …')

27 other rules: +0.41 in total (mostly absence and length terms).

Note. An original speech in which the speaker speaks for the Commission (*det förslag vi kommer att lägga fram* 'the proposal we will put forward'). Future *kommer att*, *säkerställa* 'ensure' (calque lexicon), the nominal *betänkandet av* 'the report by' and the relative clause without *som* all point to translation. Commission prose given in Swedish shares these conventions with translated Commission prose (Section 8).

The examples show the two main error sources. False positives are originals in an institutional register that shares the conventions of translated EP prose (Examples 3 and 6), and false negatives are translations in which the translator chose native variants (Example 4). In both cases, the engine's evidence is correct about the words; the words do not determine the source language.

6 Results

6.1 Classification

Table 3 and Figure 1 give accuracy and AUC on trn and val. The engine separates the classes well on 500-token chunks (val AUC .977) and moderately on single sentences (.666), where most rules have no hit. It outperforms the mean-sentence-length baseline at every length. The trn-minus-val AUC gap is .057 for sentences, .074 for 100-token chunks and .013 for 500-token chunks. Since val lies in a later parliamentary term, this gap combines overfitting of the hand calibration with drift: several markers of original Swedish become rarer from 1997 to 2004, and short texts depend on single clues more than long ones.

6.2 Family ablation

Table 4 removes one family at a time on trn. The ranking by AUC loss at 100 tokens is connectives and punctuation (−.030), register (−.025), particles (−.021), calques and the starter rules (−.011 each), word order (−.005) and statistics (−.002). The statistics family abstains below 50 words and its evidence is largely carried by other families once all are combined. Word order has a small marginal effect, spread over many weak per-sentence opening terms; it is kept because it owns the prefield evidence that was removed from other rules during integration.

6.3 Machine learning on the expert features

To measure how far the hand-set weights are from learned ones, and how much signal the expert features carry beyond the engine, we fitted logistic regression (LR), gradient-boosted trees (GBT) and shallow decision trees on trn (hyper-parameters by five-fold cross-validation inside trn, folds grouped by session date) and scored each model once on val (Table 5). The inputs are expert features only: the engine's per-rule contributions, the raw statistic each rule computes, and about 110 features from the literature word lists. An LR over the 300 most frequent words serves as a reference without expert input. None of these models is part of the engine.

Three observations follow. First, **re-weighting the hand-set rules gains little**: an LR over the 49 rule contributions changes val AUC by +.003, +.007 and −.005 at the three lengths. The learned multiplier on each hand-set contribution is positive for every rule at every length (no rule changes sign); on 100-token chunks it ranges from 0.395 to 2.385, and 43 of 49 rules lie between 0.5 and 2. The largest multipliers (sentence-initial *Men* 2.385, additive *också* 2.067, *den/det/de här* 2.017) mark rules whose hand-set weights are conservative. The smallest (*tycka* vs *anse* 0.395, *att*-drop 0.446, *ni* vs *man* 0.455) concern rules that share evidence with other rules: *tycker jag är* fires both *tycka* and *att*-drop, and every generic *man* counts in both *ni* vs *man* and the generic-*man* rate. Second, **gains come from additional features**, not from weights: models that

Table 3: Accuracy / ROC-AUC on balanced, topic-masked data (unmasked figures differ by at most .001). The engine and the baseline have no fitted parameters; trn is where the rule values were read off, val is held out. The hybrid row is the best model per column among the ML models over expert features (Section 6.3; superscript: model). Sources: `docs/INTEGRATION.md`, `hybrid/reports/METRICS.md`.

system	split	sentence	100 tokens	500 tokens
rule engine	trn	.653 / .723	.822 / .907	.945 / .990
rule engine	val	.613 / .666	.747 / .833	.920 / .977
mean sentence length	trn	.561 / .582	.629 / .677	.721 / .787
mean sentence length	val	.563 / .583	.633 / .671	.708 / .792
best expert-feature hybrid	val	.642 / .701$^{\text{gbt_all}}$	.787 / .869$^{\text{gbt_all}}$	.939 / .983$^{\text{lr_all}}$

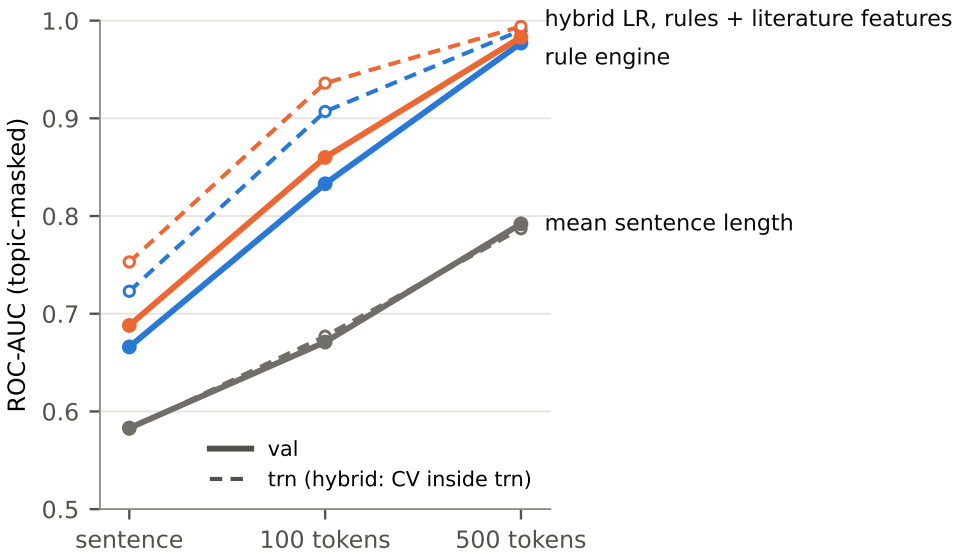


Figure 1: ROC-AUC by unit length, topic-masked. Solid: val; dashed: trn (for the hybrid model, out-of-fold AUC from cross-validation inside trn). The hybrid is a logistic regression over the engine's rule contributions, rule statistics and literature word-list features (`lr_all`). Sources: `docs/INTEGRATION.md`, `hybrid/reports/METRICS.md`.

Table 4: Leave-one-family-out ablation on trn (masked, balanced): accuracy / AUC without the family and the change (ΔAcc / ΔAUC) relative to the full engine. Source: `docs/INTEGRATION.md`.

without	sentence	Δ	100 tokens	Δ	500 tokens	Δ
all families	.653 / .723		.822 / .907		.945 / .990	
connpunct	.632 / .695	−.021 / −.028	.788 / .877	−.034 / −.030	.921 / .980	−.024 / −.010
register	.633 / .696	−.021 / −.027	.800 / .882	−.023 / −.025	.935 / .985	−.010 / −.006
particles	.636 / .702	−.018 / −.021	.801 / .886	−.021 / −.021	.939 / .984	−.006 / −.006
starter (modal particles, sent_len)	.634 / .710	−.019 / −.014	.812 / .896	−.010 / −.011	.940 / .988	−.005 / −.002
calques	.647 / .713	−.006 / −.010	.810 / .896	−.012 / −.011	.941 / .988	−.004 / −.003
wordorder	.649 / .717	−.004 / −.006	.817 / .902	−.005 / −.005	.945 / .989	+.000 / −.002
stats	.653 / .723	.000 / .000	.821 / .905	−.001 / −.002	.943 / .990	−.002 / .000

Table 5: Val AUC (topic-masked) of models fitted on trn (hyper-parameters by session-grouped CV inside trn) and scored once on val. For the bag-of-words reference, the unmasked AUC is in parentheses. Source: `hybrid/reports/METRICS.md`.

model	key	sentence	100 tokens	500 tokens
rule engine (no fitting)	`engine_raw`	.666	.833	.977
engine score, LR threshold and scale	`engine_calib`	.666	.833	.977
LR re-weighting of the 49 rule contributions	`stack_rules`	.669	.840	.972
LR over the 7 family subscores	`stack_families`	.667	.830	.974
LR over rule contributions + rule statistics	`lr_engine`	.675	.847	.972
GBT over rule contributions + rule statistics	`gbt_engine`	.681	.845	.968
LR over ∼110 literature word-list features	`lr_expert`	.670	.824	.964
GBT over the literature features	`gbt`	.684	.838	.959
engine score + 10 literature features (LR)	`hybrid_plus`	.681	.850	.981
LR over rules + statistics + literature features	`lr_all`	.688	.860	.983
GBT over rules + statistics + literature features	`gbt_all`	.701	.869	.971
LR over the 300 most frequent words (no expert input)	`bow_lr`	.700 (.707)	.870 (.880)	.971 (.975)
baseline: mean sentence length	`MeanSentLen`	.583	.671	.792

add the literature features gain .035 AUC on sentences, .036 on 100-token chunks and .006 on 500-token chunks over the engine. Third, **expert features match word frequencies once topic is masked**: the bag-of-words model is the only one that loses AUC under masking (.707 to .700 on sentences, .880 to .870 on 100-token chunks). Masked, it is level with the best expert models on sentences (.700 vs .701) and 100-token chunks (.870 vs .869) and below them on 500-token chunks (.971 vs .983).

7 Findings on Swedish translationese in Europarl

The calibration statistics give a description of EN→SV translationese in EP proceedings of 1997–2004. Unless stated otherwise, figures are hand-set rule values or calibration statistics on trn (rates per 1000 words, mostly from 500-token chunks; shares for choices), original vs translated.

Unique items are under-represented in translation, as predicted by the unique-items hypothesis (Tirkkonen-Condit 2004) and contrastive work on Swedish modal particles (Aijmer 1996). Modal particle *ju* occurs at 0.96 vs 0.054; the rules for *nämligen* (0.42 vs 0.02), *bland annat* (0.64 vs 0.021) and stance adverbs such as *egentligen* (0.51 vs 0.12) are among the strongest O evidence. Swedish-specific lexical choices behave the same way: *tycka* takes .58 of *tycka+anse* in originals and .2 in translations, *väldigt* .43 vs .1 of *väldigt+mycket* before adjectives, and generic *man* is more frequent in originals (3.5 vs 2.4).

Interference from English is visible in function words and fixed phrases. The future with *kommer att* (English *will*) takes .24 of *kommer att+ska/skall* in originals and .6 in translations; *skulle vilja* ('would like to') .097 vs .32 of *skulle vilja+vill*; the non-reflexive possessives *hans/hennes/deras/dess* occur at 0.98 vs 2.8. A comma after a fronted adverbial follows 1% of such adverbials in originals and 9.1% in translations. The preposition *av* is more frequent in translations (13.6 vs 19), in line with Nilsson (2004). Translations have longer sentences (mean 18.3 vs 21.5 words), the direction Volansky et al. (2015) report for English. Word order matches the prediction that translators keep the English theme (Hasselgård 2004; Bohnacker and Rosén 2008) only in part: topicalisation leans original when it is a Swedish-initiated choice (deictic or causal adverbs, object *det*, *Det vet vi* 'that know we'), whereas fronted adverbial clauses and structuring conjuncts (*För det första* 'firstly', *Slutligen* 'finally') lean translated, since they mirror an English sentence-initial adverbial. Subject-pronoun-first openings lean translated in 1997–2001, and the difference fades by 2002–2004.

Claims that do not hold in this corpus:

- *Explicitation through connectives* (Blum-Kulka 1986; Volansky et al. 2015). The T-leaning connectives of the catalogue are *less* frequent in translations (7.02 vs 4.96), and the native connective adverbs in the middle field are a strong O feature (4.7 vs 1.9).
- *Normalisation towards formal variants* as reported for EP Swedish by Danielsson (2003). *skall* and *ska* together occur at 6.4 vs 2.8, so the modal is an O feature; under the EP house style of the period, *skall* takes at least 98% of the two in both classes (share of *ska* .0022 vs .014). *emellertid* (0.55 vs 0.59) and *huruvida* (0.089 vs 0.13) show no usable difference, and *således* is more frequent in originals (0.13 vs 0.050). *den/det/de här*, an O keyword in that study, is more frequent in translations (0.17 vs 0.90). Formal *samt* leans original (0.85 vs 0.25), and so do *naturligtvis/givetvis/självklart* taken together (1.2 vs 0.529); only within the choice *naturligtvis* vs *givetvis/förstås* is *naturligtvis* the translated variant (.71 vs .85). Danielsson compared EP translations with debates of the Swedish Riksdag, not with EP originals, which may explain part of the difference.
- *Simplification* (Baker 1993; Laviosa 1998). Lexical variety is equal at two significant digits: MATTR over 50-word windows .830 vs .828, OVIX 64.3 vs 63.5.
- *Translations avoid sentence-initial conjunctions*, predicted by analogy with English (Volansky et al. 2015): sentence-initial *Och* (0.016 vs 0.17) and *Men* (0.45 vs 0.85) are more frequent in translations.
- Resumptive *så* after a fronted clause is not an O feature, and *Det är därför* 'that is why' leans original rather than translated.
- Most textbook calques have no translation lean in EP Swedish: *i termer av*, *på basis av*, *i ljuset av*, *baserat på* and *sist men inte minst*. *implementera* occurs almost only in originals (0.039 vs 0.0003).

Two properties of the corpus explain much of this. The originals are edited transcripts of speeches, so spoken-register items (particles, *väldigt*, *bra*, exclamations) mark them; and the translations are produced under institutional norms of EU Swedish that favour source-shaped, nominal prose (Bendegard 2014). Features that separate the classes here may therefore separate speech from institutional writing as much as originals from translations.

8 Limitations

- **One domain.** The rules were built for and tested on edited European Parliament proceedings. Some rules partly encode EP or transcription conventions and say so in their rationale (written-out abbreviations, dash-delimited asides, exclamation marks). Nothing has been tested on fiction, news or other genres, where cross-domain accuracy of translationese classifiers is known to drop (Rabinovich and Wintner 2015).
- **Drift.** Several markers of original Swedish decline over 1997–2004, and the rule values reflect that period. Val already scores below trn; test (2007–2011) is likely to score lower again.
- **Speaker mix.** The originals come from a limited set of Swedish MEPs (151 distinct speakers in trn), while the translated class is dominated by European Commission speakers. Rule hits were checked for speaker concentration, but the confound cannot be removed entirely; Example 6 shows an original that reads like translated Commission prose.
- **Short texts.** A single sentence carries little evidence: val AUC is .666 on sentences against .977 on 500-token chunks.
- **Labels.** Source-language tags are incomplete and partly inconsistent across language versions; the cross-checks drop inconsistent turns but cannot detect relay translation.
- **Test not run.** All held-out figures are from val, which was used, sparingly, to compare rule sets. The locked test split (2007–2011, 589 + 589 500-token chunks) has not been scored.

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A Rule list

Table 6: All 49 enabled rules, generated from the rule metadata (`python3 -m tsw --list-rules`). Type: hit, rate (Poisson), choice (binomial), stat (Gaussian), pred. (predicate). Dir: direction of a hit (for choice rules, of variant A). Cap: per-rule cap on $|w \cdot \mathrm{LLR}|$; w: weight.

rule	type	dir	cap	w	description
starter (2 rules, family cap 6)					
`starter.modal_particles`	rate	O	3	1.00	modal particles ju/nog/väl/visst
`starter.sent_len`	stat	T	2	0.50	words per sentence
connectives & punctuation (13 rules, family cap 4)					
`connpunct.native_connectives`	rate	O	2	0.50	native connective adverbs (dessutom, därför, dock, därmed, inte minst …)
`connpunct.additive_ocksa`	rate	O	1.5	0.40	additive 'också'
`connpunct.calque_connectives`	rate	T	2	0.80	Swedish equivalents of English discourse markers (i synnerhet, i själva verket …)
`connpunct.att_drop`	rate	O	1.5	1.00	complementizer 'att' dropped after tror/tycker/anser/hoppas/säger …
`connpunct.som_drop`	choice	T	1.5	0.70	object relative without 'som' ('det arbete ni har gjort')
`connpunct.initial_och`	rate	T	2	1.00	sentence-initial 'Och'
`connpunct.initial_men`	rate	T	1.5	0.70	sentence-initial 'Men'
`connpunct.initial_adverbial_comma`	choice	T	2	1.00	comma after a sentence-initial adverbial ('Dessutom, …')
`connpunct.clause_comma_omission`	choice	T	1.5	0.70	no comma before clausal men/eftersom/medan/vilket
`connpunct.written_out_abbrev`	choice	T	1.5	0.50	'till exempel/så kallad/till och med' written out vs 't.ex./s.k./t.o.m.'
`connpunct.parenthetical_dash`	rate	T	1.5	0.50	dash between words (parenthetical insertion)
`connpunct.explanatory_parens`	rate	T	1	1.00	explanatory parentheses (no numbers, not document codes)
`connpunct.exclamation`	pred.	O	1.5	0.50	exclamation marks outside vocatives
register (9 rules, family cap 6)					
`register.kommer_att_vs_ska`	choice	T	2.5	1.00	future/intention auxiliary 'kommer att' vs 'ska/skall'
`register.skulle_vilja_vs_vill`	choice	T	2	1.00	'jag/vi skulle vilja' vs 'jag/vi vill'
`register.tycker_vs_anser`	choice	O	2	1.00	opinion verb 'tycka' vs 'anse'
`register.valdigt_vs_mycket`	choice	O	1.5	1.00	degree adverb 'väldigt' vs 'mycket' + adjective
`register.sjalvklart`	rate	O	1.5	1.00	stance adverb 'självklart'
`register.dem_har_rate`	rate	T	1	0.50	demonstrative 'den/det/de här'
`register.bland_annat`	rate	O	1.5	0.60	'bland annat / bl.a.'
`register.samt`	rate	O	1.5	1.00	coordinator 'samt'
`register.bra`	rate	O	1.5	1.00	evaluative 'bra'
particles & pronouns (15 rules, family cap 4)					
`particles.namligen`	rate	O	2.5	1.00	clause-internal particle 'nämligen' (explanatory 'you see')

rule	type	dir	cap	w	description
`particles.anda`	rate	O	2	1.00	concessive particle ‘ändå’
`particles.alltsa`	rate	O	1.5	1.00	inferential particle ‘alltså’
`particles.stance`	rate	O	2	1.00	stance adverbs ‘egentligen/visserligen/i och för sig/i alla fall’
`particles.da_anaphoric`	rate	O	2	1.00	anaphoric/inferential ‘då’ (‘then, in that case’)
`particles.of_course`	choice	T	1.5	1.00	‘of course’: ‘naturligtvis’ vs ‘givetvis/förstås’
`particles.dvs`	rate	O	1.5	1.00	reformulation marker ‘dvs.’ / ‘det vill säga’
`particles.particle_verbs`	rate	O	1.5	1.00	Swedish-specific particle verbs (slå fast, ställa sig bakom, lyfta fram, …)
`particles.ta_itu_med`	rate	T	2	1.00	‘ta itu med’ (EN tackle/address/deal with)
`particles.generic_man`	rate	O	1.5	0.25	generic pronoun ‘man’
`particles.copular_det`	rate	O	1.5	0.50	copular ‘det är/var’ and ‘är/var det’ (expletive, cleft, anaphoric det)
`particles.poss_1st`	rate	T	1.5	0.60	1st-person possessive determiners ‘min/mitt/mina/vår/vårt/våra’
`particles.poss_3rd`	rate	T	2	0.80	3rd-person non-reflexive possessives ‘hans/hennes/deras/dess’
`particles.han_hon`	rate	T	1	0.50	3rd-person personal pronouns ‘han/hon’
`particles.ni_vs_man`	choice	T	1.5	1.00	‘ni’ vs generic ‘man’
calques (4 rules, family cap 3)					
`calques.lexicon`	hit	T	3	1.00	English calques and anglicisms
`calques.native_counterparts`	hit	O	2	1.00	native alternatives of calques
`calques.forsoka_att`	choice	T	1.5	1.00	‘försöka att + V’ vs ‘försöka + V’
`calques.pied_piped_relative`	hit	T	1.5	1.00	pied-piped relative ‘P + vilken/vilket/vilka’
word order (3 rules, family cap 4)					
`wordorder.fronted_theme`	pred.	O	3	0.80	topicalised non-subject in the prefield (V2 inversion): which kind
`wordorder.subject_theme`	pred.	T	2	0.80	subject in the prefield: pronoun / NP / det / demonstrative / cleft
`wordorder.v2_violation`	pred.	T	3	1.00	English word order leaking into Swedish (V3 / no inversion / embedded negation)
statistics (3 rules, family cap 2)					
`stats.long_words`	stat	T	1	0.50	share of long words (> 6 letters, the LIX criterion)
`stats.av_nominal`	stat	T	1	1.00	‘av’ after a definite nominalisation (‘genomförandet av’)
`stats.av_other`	stat	T	1.5	1.00	‘av’ elsewhere (of-periphrasis, by-agent)